

The model. A well-mixed population follows a unanimity q -voter rule. An external controller samples q_c agents once per round, chooses NOOP or ADVOCATE using a soft stochastic policy, and, if it advocates target A , replaces one of the q social-input slots in exactly b of the N microscopic update positions selected uniformly in advance.

1. State, timing, and variables

Simple picture. Agents update one at a time; the controller acts only once per population round. In the binary theory, A is the controller target and B the alternative. At the beginning of round k ,

$$N_k = n \in \{0, \dots, N\}, \quad x_k = n/N,$$

where n is the number of A supporters. One round contains exactly N focal-update opportunities.

Symbol	Meaning
N	population size and number of microscopic positions per round
q	number of social inputs seen by a focal agent
q_c	number of agents sensed by the controller at the round start
b	exact number of controlled positions in an advocacy round; $c = b/N$
Y_k	number of target supporters found in the controller sample
U_k	round action: NOOP or ADVOCATE
K_0, K_1	one-update kernels without / with controller advocacy
R_0, R_1	whole-round kernels under NOOP / ADVOCATE

The round protocol is

$$N_k \rightarrow Y_k \rightarrow U_k \rightarrow \underbrace{N \text{ microscopic updates}}_{\text{no new controller decision}} \rightarrow N_{k+1}.$$

If $U_k = \text{ADVOCATE}$, a set $S_k \subset \{1, \dots, N\}$ with $|S_k| = b$ is drawn uniformly *before* the round. Choosing update positions, not fixed agent identities, preserves exchangeability.

2. Population rule and microscopic kernels

Simple picture. The focal agent switches only when all q inputs unanimously support the opposite opinion. At a controlled position, the controller simply occupies one social-input slot and says A ; the focal response rule itself is unchanged.

Ordinary update:

$$K_0(n+1 | n) = \frac{N-n}{N} \frac{\binom{n}{q}}{\binom{N-1}{q}}, \quad K_0(n-1 | n) = \frac{n}{N} \frac{\binom{N-n}{q}}{\binom{N-1}{q}}.$$

The diagonal term completes the row to one. Controlled update:

$$K_1(n+1 | n) = \frac{N-n}{N} \frac{\binom{n}{q-1}}{\binom{N-1}{q-1}}, \quad K_1(n-1 | n) = 0.$$

Again the diagonal completes the row. The controlled kernel is asymmetric by design; transfer entropy does not require microscopic reversibility.

3. Controller: finite sensing and a soft round decision

Simple picture. The unanimity rule belongs to the population. The sigmoid belongs only to the controller and decides whether the budget b is activated for the next round.

The controller samples q_c agents without replacement:

$$S(y | n) = \Pr(Y_k = y | N_k = n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}.$$

Given the measured target fraction y/q_c ,

$$\Pr(U_k = \text{ADVOCATE} | Y_k = y) = \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right], \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Thus the exact probability of advocacy at true state n is

$$a_n = \Pr(\text{ADVOCATE} | n) = \sum_y S(y | n) \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right].$$

For $q_c = 2$ and $\theta = 1/2$, this simplifies exactly to

$$a(x) = \frac{1}{2} + \frac{1}{2} \tanh \left(\frac{\beta}{4} \right) (1 - 2x).$$

For the illustrative choice $\beta = 4$, $a(x) = 0.880797 - 0.761594x$.

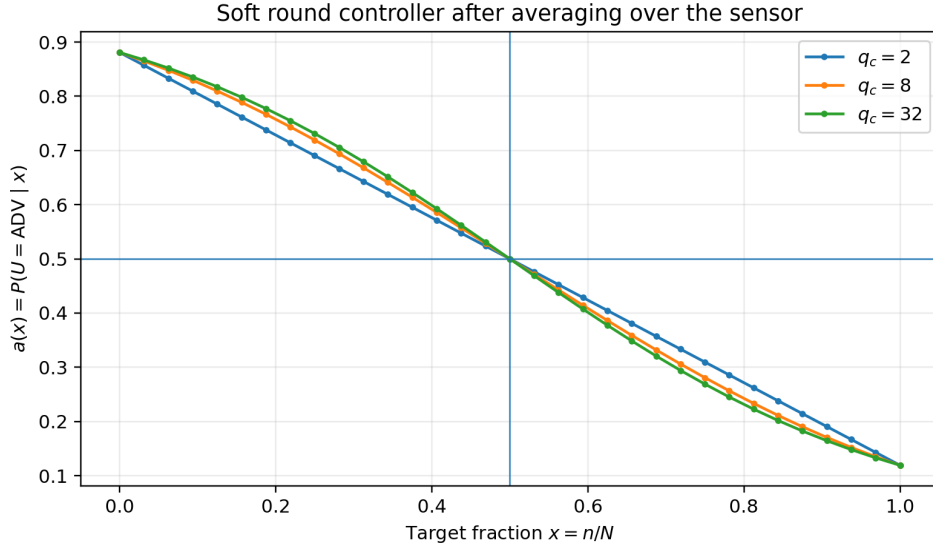


Figure 1. State-conditioned advocacy probability after averaging over the sensor for $N = 32$, $\theta = 1/2$, $\beta = 4$. q_c changes sensing fidelity; it does not change the unanimity population rule.

Parameter roles. q = social interaction order; q_c = sensing resource; θ = intervention trigger; β = controller decision stochasticity; $c = b/N$ = actuation resource.

4. Whole-round kernels

Simple picture. K_0, K_1 describe one focal update, whereas transfer entropy is measured from one round to the next. We therefore propagate the microscopic process through all N positions.

Under NOOP,

$$R_0 = K_0^N.$$

Under ADVOCATE, exactly b positions are controlled. Let $F_{r,j}$ be the transition matrix after r positions with j controlled positions already used. Starting from $F_{0,0} = I$,

$$F_{r+1,j+1} += \frac{b-j}{N-r} F_{r,j} K_1, \quad F_{r+1,j} += \frac{N-b-r+j}{N-r} F_{r,j} K_0,$$

so that

$$R_1 = F_{N,b}.$$

This is exact and averages over all uniformly preallocated control schedules without enumerating them.

5. Exact finite- N transfer entropy

Simple picture. Fix the current population state n . If knowing whether the controller chose ADVOCATE or NOOP changes the distribution of the next-round state, the action carries information into the population dynamics.

The action-marginalized next-state law is

$$M_n(m) = [1 - a_n]R_0(m | n) + a_n R_1(m | n).$$

The exact state-local transfer entropy is

$$T(n) = [1 - a_n] \sum_m R_0(m | n) \log_2 \frac{R_0(m | n)}{M_n(m)} + a_n \sum_m R_1(m | n) \log_2 \frac{R_1(m | n)}{M_n(m)}.$$

Equivalently,

$$T(n) = \text{JS}_{a_n}(R_0(\cdot | n), R_1(\cdot | n)),$$

a weighted Jensen-Shannon divergence between the exact no-control and advocacy round kernels. This is already an exact finite-population theoretical result: once $(N, q, q_c, b, \beta, \theta)$ are fixed, there is no empirical MI estimator in the calculation.

If $P_k(n)$ is the distribution of states at the start of round k ,

$$\mathcal{T}_k = I(U_k; N_{k+1} | N_k) = \sum_n P_k(n) T(n).$$

For a K -round experiment, the natural finite-horizon rate is

$$\overline{\mathcal{T}}^{(K)} = K^{-1} \sum_{k=0}^{K-1} \mathcal{T}_k.$$

The pure unanimity voter has absorbing consensus states, so long-time stationary TE can vanish after consensus. For comparison with finite LLM episodes, $T(n)$, $\mathcal{T}_k$, and $\overline{\mathcal{T}}^{(K)}$ are therefore more informative.

Two ingredients are required for positive TE. The soft policy provides action variation at fixed state through $0 < a_n < 1$. The population kernels provide dynamical separation through $R_0 \neq R_1$. If either ingredient disappears, $T(n) = 0$.

6. Closed large- N approximation

Simple picture. To understand how control should scale with q and b/N , approximate the count fraction $x = n/N$ as continuous and compare the mean drift with and without advocacy.

The ordinary and controlled drifts are

$$f_0(x) = (1 - x)x^q - x(1 - x)^q, \quad f_1(x) = (1 - x)x^{q-1}.$$

Their difference is

$$\Delta f_q(x) = x^{q-1}(1 - x)^2 + x(1 - x)^q.$$

With $c = b/N$, an advocacy round changes the mean drift by approximately $c\Delta f_q(x)$. The ordinary local noise coefficient is

$$\nu_0(x) = (1 - x)x^q + x(1 - x)^q - f_0(x)^2.$$

A second-order Jensen-Shannon expansion gives

$$T_{\text{MF}}(x) \approx \frac{a(x)[1 - a(x)]}{2 \ln 2} \frac{Nc^2[\Delta f_q(x)]^2}{\nu_0(x)}.$$

Hence the characteristic weak-control scaling is $T_{\text{MF}} \propto a(1 - a)b^2/N$, modulated by the q -dependent response-to-noise ratio.

7. Exact curves: what the classical theory predicts

Simple picture. The following are theory curves, not simulations. They use the exact $K_0, K_1 \rightarrow R_0, R_1 \rightarrow T(n)$ construction for $N = 32$, $q_c = 2$, $\theta = 1/2$, $\beta = 4$, while varying q and the round budget b .

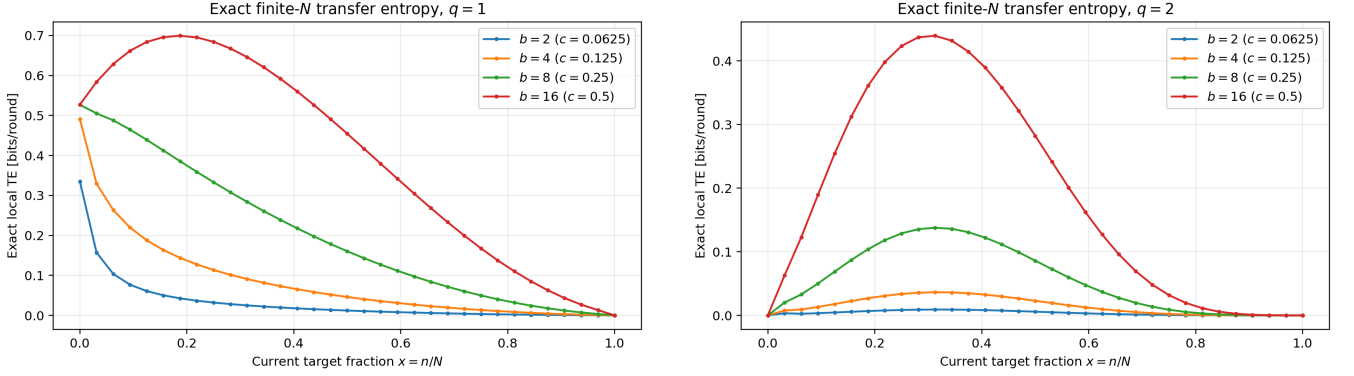


Figure 2. Exact state-local TE. For $q = 1$ (left), advocacy can seed target support even from $x = 0$, so the boundary channel can be strong. For $q = 2$ (right), $T(0) = 0$: one controller input still requires one ordinary A supporter to complete unanimity. The information channel therefore peaks in the interior.

8. The analytically stronger special case $q = 1$

For $q = 1$, ordinary copying has zero mean drift, $f_0(x) = 0$, whereas a controlled position gives $f_1(x) = 1 - x$. With exactly b controlled positions,

$$\mathbb{E}[x_{k+1} \mid x_k = x, \text{ADVOCATE}] = 1 - (1 - x) \left(1 - \frac{1}{N}\right)^b,$$

while $\mathbb{E}[x_{k+1} \mid x_k = x, \text{NOOP}] = x$. Therefore

$$\Delta\mu_{q=1}(x) = (1 - x) \left[1 - \left(1 - \frac{1}{N}\right)^b \right] \rightarrow (1 - x)(1 - e^{-c})$$

when $N \rightarrow \infty$ with $b = cN$.

This creates a qualitative boundary distinction:

$$q = 1 : T(0) > 0 \text{ for } b > 0, \quad q \geq 2 : T(0) = 0.$$

9. Control-budget dependence and compact comparison sheet

Simple picture. The final theoretical question is how much information can be injected as the round budget $c = b/N$ increases. The exact theory predicts a small-budget quadratic regime for $q \geq 2$, followed by saturation as the two round kernels become increasingly distinguishable.

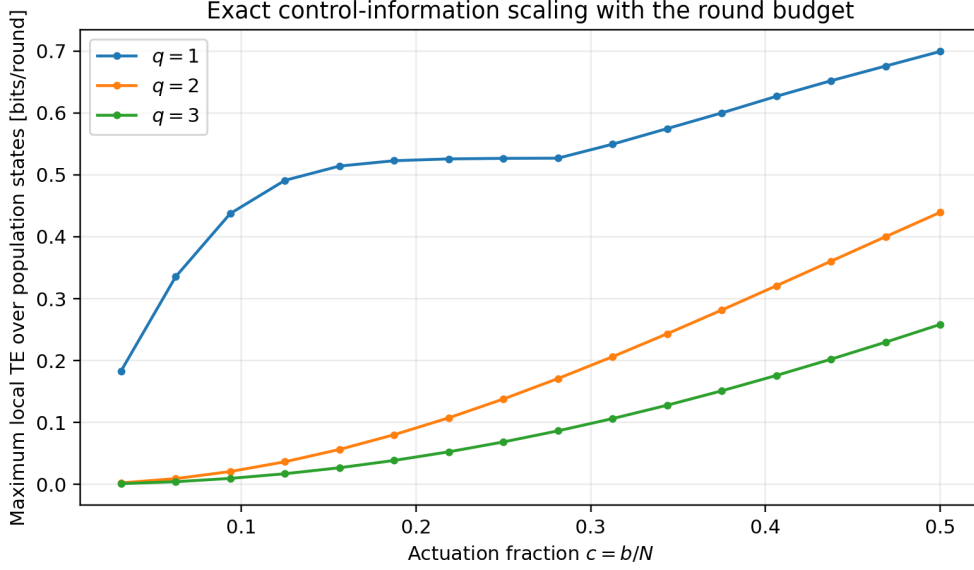


Figure 3. Maximum exact state-local TE over population states versus $c = b/N$ for $N = 32$, $q_c = 2$, $\theta = 1/2$, $\beta = 4$. For $q \geq 2$ the low-budget behavior is consistent with the c^2 scaling of the mean-field approximation. $q = 1$ departs earlier because advocacy opens otherwise inaccessible boundary transitions.

Object	Theoretical expression
Sensor	$S(y n) = \binom{n}{y} \binom{N-n}{q_c-y} / \binom{N}{q_c}$
Soft controller	$a_n = \sum_y S(y n) \sigma[\beta(\theta - y/q_c)]$
Population response	unanimity q -voter, encoded by K_0, K_1
Round dynamics	$R_0 = K_0^N, R_1 = F_{N,b}$
Exact local TE	$T(n) = \text{JS}_{a_n}[R_0(\cdot n), R_1(\cdot n)]$
Episode/round TE	$\mathcal{T}_k = \sum_n P_k(n) T(n)$
Mean-field separation	$\Delta f_q = x^{q-1}(1-x)^2 + x(1-x)^q$
Weak-control TE	$T_{\text{MF}} \approx \frac{a(1-a)}{2 \ln 2} \frac{N c^2 \Delta f_q^2}{\nu_0}$

What is fixed in this benchmark. Population response = unanimity q -voter; sensing = hypergeometric sample of size q_c ; controller decision = soft sigmoid; timing = once per round; actuation = exactly b preallocated positions, each replacing one of the q social inputs with advocacy for A . The LLM simulations can now be compared against these classical predictions at matched $(N, q, q_c, b, \beta, \theta)$.

Theory anchors. Irisarri et al., *Stochastic Thermodynamics of Social Imitation beyond Energetics* (2026), motivates the nonlinear imitation mechanism. Horowitz and Sandberg, *Second-law-like inequalities with information and their interpretations* (2014), motivates directional information in repeated feedback. The controlled kernel here is not constrained to be microscopically reversible.